

QuantCourse

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1 PDE: deriving the PDE for the contract

Derivation of the pricing PDE in integral form

Assume that the two stock prices follow independent geometric Brownian motions:

$$dS_1(t) = \mu_1 S_1(t) dt + \sigma S_1(t) dW_1(t), \quad dS_2(t) = \mu_2 S_2(t) dt + \sigma S_2(t) dW_2(t),$$

where W_1 and W_2 are independent Brownian motions. Let the derivative price be

$$V(t, S_1(t), S_2(t)),$$

where $V \in C^{1,2,2}$. Applying the two-dimensional Itô formula

$$\begin{aligned} V(t, S_1(t), S_2(t)) &= V(0, S_1(0), S_2(0)) + \int_0^t V_t(u, S_1(u), S_2(u)) du \\ &\quad + \int_0^t V_{S_1}(u, S_1(u), S_2(u)) dS_1(u) + \int_0^t V_{S_2}(u, S_1(u), S_2(u)) dS_2(u) \\ &\quad + \frac{1}{2} \int_0^t V_{S_1 S_1}(u, S_1(u), S_2(u)) d[S_1](u) \\ &\quad + \frac{1}{2} \int_0^t V_{S_2 S_2}(u, S_1(u), S_2(u)) d[S_2](u) \\ &\quad + \int_0^t V_{S_1 S_2}(u, S_1(u), S_2(u)) d[S_1, S_2](u). \end{aligned}$$

Now, since

$$dS_1(u) = \mu_1 S_1(u) du + \sigma S_1(u) dW_1(u), \quad dS_2(u) = \mu_2 S_2(u) du + \sigma S_2(u) dW_2(u),$$

their quadratic variations are

$$d[S_1](u) = \sigma^2 S_1(u)^2 du, \quad d[S_2](u) = \sigma^2 S_2(u)^2 du.$$

Because W_1 and W_2 are independent, the cross-variation is 0:

$$d[S_1, S_2](u) = 0.$$

Therefore,

$$\begin{aligned}
V(t, S_1(t), S_2(t)) &= V(0, S_1(0), S_2(0)) + \int_0^t V_t(u, S_1(u), S_2(u)) du \\
&+ \int_0^t V_{S_1}(u, S_1(u), S_2(u)) dS_1(u) + \int_0^t V_{S_2}(u, S_1(u), S_2(u)) dS_2(u) \\
&+ \frac{1}{2} \int_0^t \sigma^2 S_1(u)^2 V_{S_1 S_1}(u, S_1(u), S_2(u)) du \\
&+ \frac{1}{2} \int_0^t \sigma^2 S_2(u)^2 V_{S_2 S_2}(u, S_1(u), S_2(u)) du.
\end{aligned}$$

Substituting $dS_1(u)$ and $dS_2(u)$, we get

$$\begin{aligned}
V(t, S_1(t), S_2(t)) &= V(0, S_1(0), S_2(0)) \\
&+ \int_0^t \left(V_t(u, S_1(u), S_2(u)) + \mu_1 S_1(u) V_{S_1}(u, S_1(u), S_2(u)) \right. \\
&\quad + \mu_2 S_2(u) V_{S_2}(u, S_1(u), S_2(u)) + \frac{1}{2} \sigma^2 S_1(u)^2 V_{S_1 S_1}(u, S_1(u), S_2(u)) \\
&\quad \left. + \frac{1}{2} \sigma^2 S_2(u)^2 V_{S_2 S_2}(u, S_1(u), S_2(u)) \right) du \\
&+ \int_0^t \sigma S_1(u) V_{S_1}(u, S_1(u), S_2(u)) dW_1(u) \\
&+ \int_0^t \sigma S_2(u) V_{S_2}(u, S_1(u), S_2(u)) dW_2(u).
\end{aligned}$$

Next, define the self-financing portfolio

$$\Pi(t) = V(t, S_1(t), S_2(t)) - \Delta_1(t)S_1(t) - \Delta_2(t)S_2(t).$$

Its dynamics in integral form are

$$\begin{aligned}
\Pi(t) - \Pi(0) &= V(t, S_1(t), S_2(t)) - V(0, S_1(0), S_2(0)) \\
&- \int_0^t \Delta_1(u) dS_1(u) - \int_0^t \Delta_2(u) dS_2(u).
\end{aligned}$$

Substituting the expression for $V(t, S_1(t), S_2(t))$, we obtain

$$\begin{aligned}
\Pi(t) - \Pi(0) &= \int_0^t \left(V_t + \mu_1 S_1 V_{S_1} + \mu_2 S_2 V_{S_2} + \frac{1}{2} \sigma^2 S_1^2 V_{S_1 S_1} + \frac{1}{2} \sigma^2 S_2^2 V_{S_2 S_2} \right) (u, S_1(u), S_2(u)) du \\
&+ \int_0^t \sigma S_1(u) V_{S_1}(u, S_1(u), S_2(u)) dW_1(u) \\
&+ \int_0^t \sigma S_2(u) V_{S_2}(u, S_1(u), S_2(u)) dW_2(u) \\
&- \int_0^t \Delta_1(u) dS_1(u) - \int_0^t \Delta_2(u) dS_2(u).
\end{aligned}$$

Choose the hedge ratios

$$\Delta_1(u) = V_{S_1}(u, S_1(u), S_2(u)), \quad \Delta_2(u) = V_{S_2}(u, S_1(u), S_2(u)).$$

Then the stochastic terms cancel, and the portfolio becomes locally riskless:

$$\Pi(t) - \Pi(0) = \int_0^t \left(V_t + \frac{1}{2} \sigma^2 S_1^2 V_{S_1 S_1} + \frac{1}{2} \sigma^2 S_2^2 V_{S_2 S_2} \right) (u, S_1(u), S_2(u)) du.$$

Since the portfolio is riskless, by the no-arbitrage principle it must earn the risk-free rate:

$$\Pi(t) - \Pi(0) = \int_0^t r\Pi(u) du.$$

Using

$$\Pi(u) = V(u, S_1(u), S_2(u)) - S_1(u)V_{S_1}(u, S_1(u), S_2(u)) - S_2(u)V_{S_2}(u, S_1(u), S_2(u)),$$

we get

$$\begin{aligned} \Pi(t) - \Pi(0) = \int_0^t r \Big(& V(u, S_1(u), S_2(u)) - S_1(u)V_{S_1}(u, S_1(u), S_2(u)) \\ & - S_2(u)V_{S_2}(u, S_1(u), S_2(u)) \Big) du. \end{aligned}$$

Equating the two expressions for $\Pi(t) - \Pi(0)$, we obtain

$$\begin{aligned} & \int_0^t \left(V_t + \frac{1}{2}\sigma^2 S_1^2 V_{S_1 S_1} + \frac{1}{2}\sigma^2 S_2^2 V_{S_2 S_2} \right) (u, S_1(u), S_2(u)) du \\ & = \int_0^t r \left(V - S_1 V_{S_1} - S_2 V_{S_2} \right) (u, S_1(u), S_2(u)) du. \end{aligned}$$

Hence, the integrands must be equal, and therefore the price function satisfies

$$V_t + \frac{1}{2}\sigma^2 S_1^2 V_{S_1 S_1} + \frac{1}{2}\sigma^2 S_2^2 V_{S_2 S_2} + rS_1 V_{S_1} + rS_2 V_{S_2} - rV = 0.$$

The terminal condition is

$$V(T, S_1, S_2) = \max\{S_1, S_2\}.$$

2 The numerical scheme

$$V_t(x_i, y_j, t_n) \approx \frac{V_{i,j}^n - V_{i,j}^{n-1}}{\Delta t}$$

$$V_t = F$$

Substituting

$$V_{i,j}^{n-1} \approx V_{i,j}^n - \Delta t F_{i,j}^n$$

$$V^{n-1} \approx V^n + \Delta t (aV_x^n + aV_y^n + bV_{xx}^n + bV_{yy}^n - rV^n)$$

where

$$a = r - \frac{\sigma^2}{2}, \quad b = \frac{\sigma^2}{2}.$$

Approximating the partial derivatives

$$V_x(x_i, y_j, t_n) \approx \frac{V_{i+1,j}^n - V_{i-1,j}^n}{2\Delta x}$$

$$V_y(x_i, y_j, t_n) \approx \frac{V_{i,j+1}^n - V_{i,j-1}^n}{2\Delta y}$$

$$V_{xx}(x_i, y_j, t_n) \approx \frac{V_{i+1,j}^n - 2V_{i,j}^n + V_{i-1,j}^n}{\Delta x^2}$$

$$V_{yy}(x_i, y_j, t_n) \approx \frac{V_{i,j+1}^n - 2V_{i,j}^n + V_{i,j-1}^n}{\Delta y^2}$$